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Standard Practice for Design and Manufacture of Turbine Engines for Unmanned Aircraft System[1]

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1. Scope

1.1 This practice covers minimum requirements for the design and manufacture of turbine engines for unmanned aircraft systems.

1.2 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.*

2. Significance and Use

2.1 This practice provides designers and manufacturers of turbine engines for Unmanned Aircraft Systems design references and criteria to use in designing and manufacturing engines.

2.2 Declaration of compliance is based on testing and documentation during the design and testing or flight testing of the engine type by the manufacturer or under the manufacturers' guidance.

3. Engine Model Designation

3.1 *Engine Parts List*—A parts list is required for each engine model qualified in accordance with this specification.

3.2 *New Engine Model Designations:*

3.2.1 Each new engine model must be qualified in accordance with this practice.

3.2.2 Design or configuration changes that impact the installation interface, performance, or operability of the engine require a new engine model designation.

3.3 *Design Changes of Parts*—Each design change of a part or component of an engine model qualified to this specification should be evaluated relative to the requirements of this specification.

4. Data Requirements

4.1 *Retained Data*—The following data and information shall be retained on file at the manufacturer's facility for at a minimum of 18 years after production is discontinued.

4.1.1 Drawings that define the engine configuration.

4.1.2 Material and process specifications referenced in the parts drawings.

4.1.3 Engineering analyses and test data prepared for qualification with this specification.

4.2 *Delivered Data*—The following data shall be delivered to the airplane manufacturer to support design and operation of the applicable airplane.

4.2.1 An engine performance specification that defines the engine performance under all anticipated operating environments.

4.2.2 An installation manual that defines all functional and physical interface requirements of the engine. This shall include an engine outline/installation drawing.

4.2.3 An operating manual that defines normal and abnormal operating procedures and any applicable operating limitations.

4.2.4 A maintenance manual that defines periodic installed maintenance, major inspection, overhaul intervals, and any other maintenance limitations.

4.2.5 An overhaul manual that provides instruction for disassembling the engine to replace or repair, or both, parts as required to return the engine to airworthy condition that is safe for operation until the next major overhaul.

5. Design Criteria

5.1 *Materials*—The materials used in the engine must be adequate for the intended design conditions of the engine.

5.2 *Fire Prevention*—The design and construction of the engine and the materials used must minimize the probability of the occurrence and spread of fire by:

5.2.1 Using fire-resistant lines, fittings, and other components that contain a flammable liquid when supplied with the engine; and

5.2.2 Shielding or locating components to safeguard against the ignition of leaking flammable fluid.

[1] This practice is under the jurisdiction of ASTM Committee F38 on Unmanned Aircraft Systems and is the direct responsibility of Subcommittee F38.01 on Airworthiness.

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5.3 *Engine Cooling*—The engine design must include provisions for cooling; the installation manual must specify engine and component temperature limitations.

5.4 *Engine Mounting*—Attach points on the engine must have data for the correct design of mounting structures to the airframe. The maximum allowable limit and ultimate loads for the engine mounting attachments and related structure must be specified.

5.5 *Ignition Systems*—Each turbine engine must have a dual-ignitor system with at least two ignitors for each engine and two separate electric circuits with separate sources of electrical energy, or have an ignition system of equivalent in-flight reliability. System maintenance must be specified under the maintenance manual supplied by the manufacturer.

5.5.1 Engines used in aircraft where the engine is not required for flight, for example, gliders, can be single ignition.

5.6 *Electronic Engine Controllers (EEC):*

5.6.1 *Single Fault Tolerance*—The EEC should be designed to accommodate single failures of the electrical circuit. Loss of any single EEC should not cause significant power reduction or engine stoppage.

5.6.2 The functioning of EEC's parameter must not be adversely affected by the declared environmental conditions of operation by the manufacturer, including temperature and moisture. The limits to which the system has been qualified shall be documented in the installation manual. For protection against radiated EMI/HIRF, the harnesses or cables should be shielded from each sensor to each end point and electrically bonded to the engine. Filter pin connectors should be located at the controller housing interface and shunted to ground on the case. Filter pin connectors should have 40 dB attenuation, minimum. For EMI emissions, powerline filters suppress emissions from the controller on outgoing signals.

5.7 *Fuel and Induction System:*

5.7.1 *Induction System Icing*—The fuel and air intake passages must be designed to minimize the accretion of ice.

5.7.2 *Filtering*—The type and degree of fuel and air filtering necessary to prevent obstruction of air or fuel flow must be specified.

5.7.3 *Liquid Lock*—Each passage in the induction system that conducts a mixture of fuel and air must be self-draining or demonstrated to not cause damage from hydraulic lock on starting.

5.8 *Lubrication System:*

5.8.1 The lubrication system of the engine must be designed and constructed so that it will function properly in all flight attitudes and atmospheric conditions in which the engine is expected to operate. In wet sump engines, this requirement must be met when only one-half of the maximum lubricant supply is in the engine.

5.8.2 The lubrication system of the engine must be designed and constructed to allow installing a means of cooling the lubricant if required.

5.8.3 The oil system must be vented to the atmosphere to preclude leakage of oil from excessive pressure in the system. This venting must have a means to prevent the blockage of the vent by ice.

5.9 *Vibration General*—The turbine engine must be designed and constructed to function throughout its normal operating range of rotational speeds and engine powers without inducing excessive stress in any of the engine parts.

5.9.1 The engine must have a rotational vibration survey to determine torsional and bending characteristics from idle speed up to maximum desired takeoff speed. This survey should be done with a representative propeller attached to the gearbox, if installed, and no hazardous conditions would be allowed.

6. Qualification Tests

6.1 *Calibration Test*—Each engine design shall be tested and the characteristics of engine rated power, speeds, and fuel consumption shall be determined.

6.2 *Durability Testing*—Each engine model must be subjected to an engine test that will verify durability by one of the following methods.

6.2.1 *Accelerated Overhaul Test*—This test simulates an engine overhaul interval. A protocol for this test shall incorporate, as a minimum, the following elements:

6.2.1.1 At least 100 % of the time at maximum power that would occur over the overhaul interval.

NOTE 1—For calculation, each hour of normal flight would have 5 min of full power.

6.2.1.2 At least 10 % of the time at cruise power that would occur over the overhaul interval.

6.2.1.3 At least one cycle per hour of test from maximum power to cruise power and back.

6.2.1.4 At least one engine start for each 5 h of testing.

6.2.1.5 Maximum engine operating temperature or maximum oil temperature, or both, shall not be exceeded.

6.2.1.6 The engine, if gearbox equipped, must be fitted with a propeller/rotor that thrust-loads the engine to the maximum thrust that the engine is designed to resist at each applicable operating condition specified in this section.

6.2.1.7 Each accessory drive and mounting attachment must be loaded. During operation at maximum power, the load imposed by each accessory used only for an aircraft service must be the limit load specified by the applicant for the engine drive or attachment point.

6.2.1.8 After completing the accelerated overhaul test, each engine must be completely disassembled and each component must conform to the new or overhaul limits established by the designer/manufacturer.

6.2.2 *Endurance Testing by Fleet Leader*—In place of the accelerated overhaul test in 6.2.1, the engine may complete endurance flight testing.

6.2.2.1 The Fleet Leader Test Method is the operation of the make and model of the engine being developed on a flying aircraft under the control of the engine manufacturer.

6.2.2.2 All engine maintenance must be documented on the engine, and no components may be replaced on the engine during the test. If components are changed, the test must restart at 0 h.

6.2.2.3 Periodic inspection must be performed. The flight tests shall subject the engine to all atmospheric conditions for which the manufacture states the engine may operate under in the operating manual required in 4.2.3.

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NOTE 2—Atmospheric conditions in 6.2.2.3 are intended to cover outside air temperatures, density altitudes, and humidity.

6.2.2.4 There must be at least one engine start for each 5 h of operation.

6.2.2.5 The engine, if gearbox equipped, must be fitted with a propeller/rotor that thrust-loads the engine to the maximum thrust that the engine is designed to resist at full power.

6.2.2.6 Each accessory drive and mounting attachment must be loaded. The load imposed by each accessory used only for an aircraft service must be the limit load specified by the applicant for the engine drive of attachment point.

6.2.2.7 After completing the Endurance Test, each engine must be completely disassembled and each component must conform to the new or overhaul limits established by the designer/manufacturer.

6.3 *Engine Overhaul Interval*—The engine overhaul interval shall be reported in the operations manual (see 4.2.3) as either the overhaul time used to complete the accelerated overhaul test in 6.2.1 or 80 % of the time accumulated on the engine model fleet leader method from 6.2.2.

NOTE 3—These engines are available for use on all sizes of unmanned aircraft systems. This practice does not restrict the use and selection of FAA certificated engines for use on unmanned aircraft systems, but offers suitable alternatives.

7. Manufacturing Requirements

7.1 The engine manufacturer shall establish inspections and tests necessary to ensure that each article produced conforms to the design and is in a condition for safe operation, including as applicable:

7.1.1 Inspections for raw materials, purchased items, and parts and assemblies produced by suppliers, including methods used to ensure acceptable quality of parts and assemblies that cannot be completely inspected for conformity and quality when delivered to the engine manufacturer's facility.

7.1.2 Production inspection of individual parts and complete assemblies, including the identification of any special manufacturing processes involved, the means used to control the processes, and the final test procedure for the completed engine.

7.1.3 A nonconforming materials review system that includes documentation of parts disposition decisions, and a system to dispose of rejected parts.

7.1.4 A system for informing company inspectors of current changes in engineering drawings, specifications, and quality control procedures.

8. Keywords

8.1 turbine engine; unmanned aircraft system

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